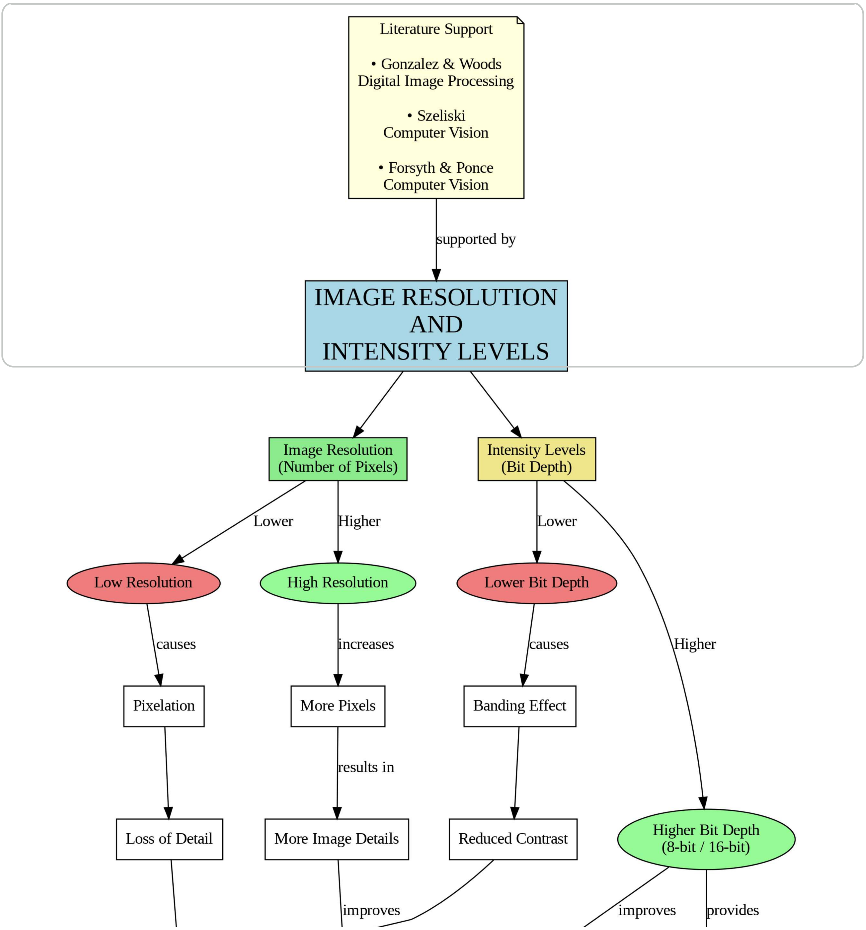
```
# ============================================================
# CONCEPT MAP: IMAGE RESOLUTION & INTENSITY LEVELS
# Google Colab Code
# ============================================================

# Install Graphviz
!apt-get -qq install graphviz
!pip -q install graphviz

from graphviz import Digraph
from IPython.display import Image, display

# Create the concept map
dot = Digraph('ConceptMap', format='png')

# Layout Settings
dot.attr(rankdir='TB')          # Top to Bottom
dot.attr(size='14,14')
dot.attr(dpi='300')
dot.attr(nodesep='0.5')
dot.attr(ranksep='0.8')

# ----------------------------
# Main Topic
# ----------------------------
dot.node('A',
         'IMAGE RESOLUTION\nAND\nINTENSITY LEVELS',
         shape='box',
         style='filled',
         fillcolor='lightblue',
         fontsize='22')

# ----------------------------
# Resolution Section
# ----------------------------
dot.node('B',
         'Image Resolution\n(Number of Pixels)',
         shape='box',
         style='filled',
         fillcolor='lightgreen')

dot.node('C',
         'High Resolution',
         shape='ellipse',
         style='filled',
         fillcolor='palegreen')

dot.node('D',
         'Low Resolution',
         shape='ellipse',
         style='filled',
         fillcolor='lightcoral')

dot.node('E',
         'More Pixels'
```

```
         shape='box')

dot.node('F',
         'More Image Details',
         shape='box')

dot.node('G',
         'Sharper Edges',
         shape='box')

dot.node('H',
         'Pixelation',
         shape='box')

dot.node('I',
         'Loss of Detail',
         shape='box')

# ----------------------------
# Intensity Section
# ----------------------------
dot.node('J',
         'Intensity Levels\n(Bit Depth)',
         shape='box',
         style='filled',
         fillcolor='khaki')

dot.node('K',
         'Higher Bit Depth\n(8-bit / 16-bit)',
         shape='ellipse',
         style='filled',
         fillcolor='palegreen')

dot.node('L',
         'Lower Bit Depth',
         shape='ellipse',
         style='filled',
         fillcolor='lightcoral')

dot.node('M',
         'Smooth Brightness\nTransitions',
         shape='box')

dot.node('N',
         'High Dynamic Range',
         shape='box')

dot.node('O',
         'Banding Effect',
         shape='box')

dot.node('P',
         'Reduced Contrast',
         shape='box')

# ----------------------------
```

```
# Image Quality
# ----------------------------
dot.node('Q',
         'High Image Quality',
         shape='box',
         style='filled',
         fillcolor='lightskyblue')

dot.node('R',
         'Low Image Quality',
         shape='box',
         style='filled',
         fillcolor='salmon')

# ----------------------------
# Usability
# ----------------------------
dot.node('S',
         'Improved Usability',
         shape='box',
         style='filled',
         fillcolor='lightcyan')

dot.node('T',
         'Reduced Usability',
         shape='box',
         style='filled',
         fillcolor='mistyrose')

dot.node('U',
         'Medical Imaging',
         shape='box')

dot.node('V',
         'Remote Sensing',
         shape='box')

dot.node('W',
         'Face Recognition',
         shape='box')

dot.node('X',
         'Autonomous Driving',
         shape='box')

# ----------------------------
# Literature Support
# ----------------------------
dot.node('Y',
         'Literature Support\n\n'
         '• Gonzalez & Woods\n'
         'Digital Image Processing\n\n'
         '• Szeliski\n'
         'Computer Vision\n\n'
         '• Forsyth & Ponce\n'
         'Computer Vision',
         shape='note',
```

```
         style='filled',
         fillcolor='lightyellow')

# ----------------------------
# Connections
# ----------------------------

# Main branches
dot.edge('A','B')
dot.edge('A','J')

# Resolution
dot.edge('B','C',label='Higher')
dot.edge('B','D',label='Lower')

dot.edge('C','E',label='increases')
dot.edge('E','F',label='results in')
dot.edge('F','G',label='improves')
dot.edge('G','Q')

dot.edge('D','H',label='causes')
dot.edge('H','I')
dot.edge('I','R')

# Intensity
dot.edge('J','K',label='Higher')
dot.edge('J','L',label='Lower')

dot.edge('K','M',label='provides')
dot.edge('K','N',label='improves')
dot.edge('M','Q')
dot.edge('N','Q')

dot.edge('L','O',label='causes')
dot.edge('O','P')
dot.edge('P','R')

# Image Quality
dot.edge('Q','S',label='leads to')
dot.edge('R','T',label='leads to')

# Applications
dot.edge('S','U')
dot.edge('S','V')
dot.edge('S','W')
dot.edge('S','X')

# Literature
dot.edge('Y','A',label='supported by')

# Save and display
filename = dot.render('Image_Resolution_Intensity_Concept_Map')

display(Image(filename))

print("Concept Map generated successfully!")
```


Literature Support
• Gonzalez & Woods
Digital Image Processing
• Szeliski
Computer Vision
• Forsyth & Ponce
Computer Vision
supported by
IMAGE RESOLUTION
AND
INTENSITY LEVELS
Image Resolution
(Number of Pixels)
Intensity Levels
(Bit Depth)
Lower
Higher
Lower
Low Resolution
High Resolution
Lower Bit Depth
causes
increases
causes
Higher
Pixelation
More Pixels
Banding Effect
results in
Loss of Detail
More Image Details
Reduced Contrast
Higher Bit Depth
(8-bit / 16-bit)
improves
improves
provides